

BMS_SLAVE Fabrication Document

Layer Stack Legend

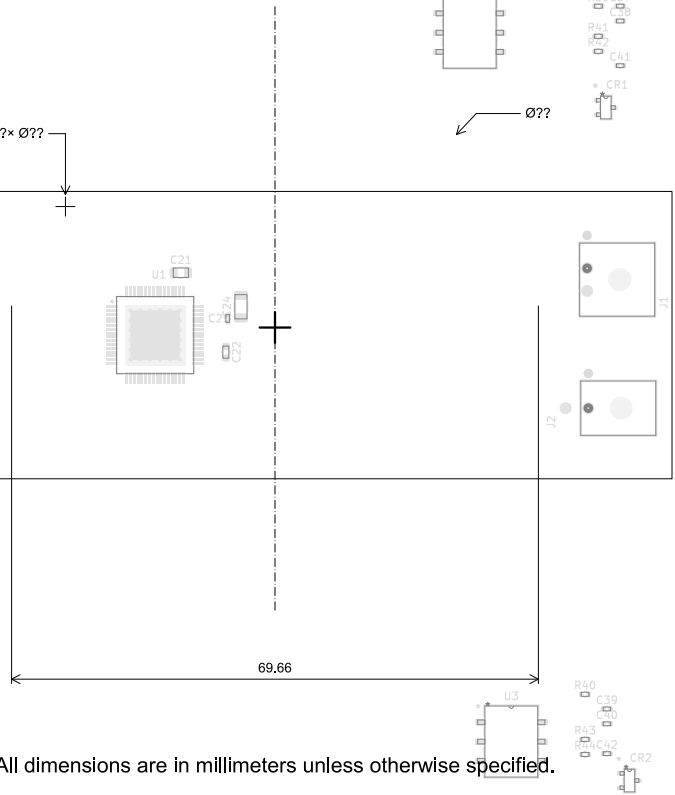
Material	Layer	Thickness	Dielectric	Type	Gerber
F,Paste				Paste Mask	
F,Silkscreen			Direct Printing	Legend	GBR
F,Mask		0,02mm	Solder Resist	Solder Mask	GBR
Copper	L1 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	L2 (GND)	0,035mm (1,00oz)		Signal	GBR
Core		1,209994mm	FR4	Dielectric	
Copper	L3 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	L4 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
B,Mask		0,02mm	Solder Resist	Solder Mask	GBR
B,Silkscreen			Direct Printing	Legend	GBR
B,Paste				Paste Mask	

Total thickness: 1,659994mm
Note: external layer thicknesses are specified after plating

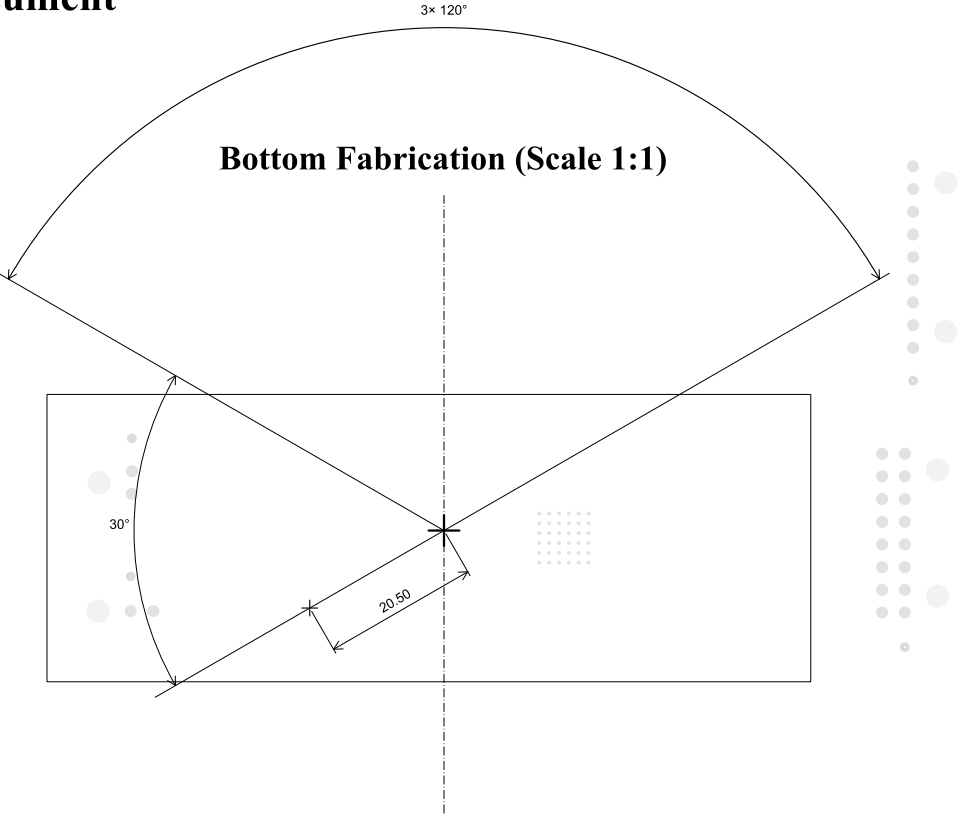
Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0,2032	0,28	L2

Top Fabrication (Scale 1:1)



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All dimensions are in millimeters unless otherwise specified.

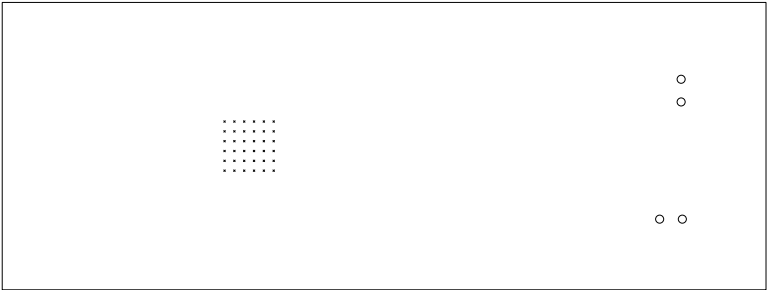
	Comments:		Company:		Variant:		Git Hash:		
			Blue Devil Motor Sports		PRELIMINARY		d2b8cb1		
			Board Name:			Project Name:			
			BMS_SLAVE			BMS Slave			
Sheet Title:		File Name:		Designer:		Date:		Revision:	
Bottom Fabrication (Scale 1:1)		bms_slave.kicad_pcb		Aidan Brzezinski		2024-04-13		+ (Unreleased)	
Sheet Path:				Reviewer:		Size:		Sheet:	
						A4		2 of 10	

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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	36	0.20mm (8,00mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
O	29	1.07mm (42,00mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
	Total 65					

Drill Drawing L1 - L4 (Scale 1:1)





Comments:

Sheet Title:
Drill Drawing (L1 - L4)

Sheet Path:

Company:
Blue Devil Motor Sports

Board Name:
BMS_SLAVE

File Name:
bms_slave.kicad_pcb

Designer:
Aidan Brzezinski

Reviewer:

Variant:
PRELIMINARY

Project Name:
BMS Slave

Date:
2024-04-13

Size:
A4

Git Hash:
d2b8cb1

Revision:
+ (Unreleased)

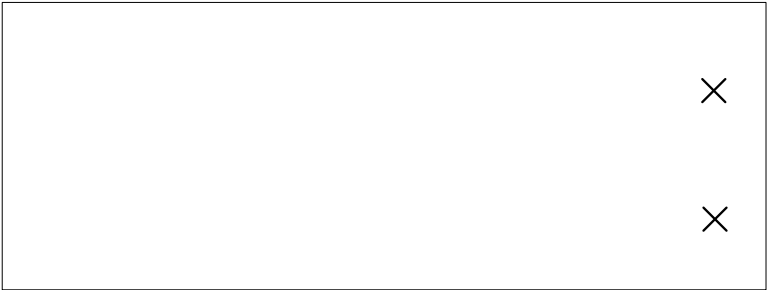
Sheet:
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	6	3.05mm (120,00mils)	NPTH	Round	L1 (Sig. PWR) - L4 (Sig. PWR)	Mechanical
	Total 6					

Drill Drawing L1 - L4 (Scale 1:1)



Comments: Sheet Title: Drill Drawing (L1 - L4) Sheet Path:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY		Git Hash: d2b8cb1
	Board Name: BMS_SLAVE		Project Name: BMS Slave		
	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)	
	Reviewer:		Size: A4	Sheet: 4 of 10	

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Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP1	CVDD	31.00	-37.05
TP2	DVDD	31.00	-41.10
TP3	AVDD	35.05	-37.05
TP4	LDOIN	35.05	-41.10
TP5	VCOLL	39.10	-37.05
TP6	LDOIN	39.10	-41.10
TP7	TSREF	43.15	-37.05
TP8	GND	43.15	-41.10
TP9	T_P1	31.00	-26.95
TP10	GND	31.00	-31.00
TP11	U1_COMLP	24.95	-26.95
TP12	U1_COMLN	24.95	-31.00
TP13	U1_COMHP	24.95	-35.05
TP14	U1_COMHN	24.95	-39.10

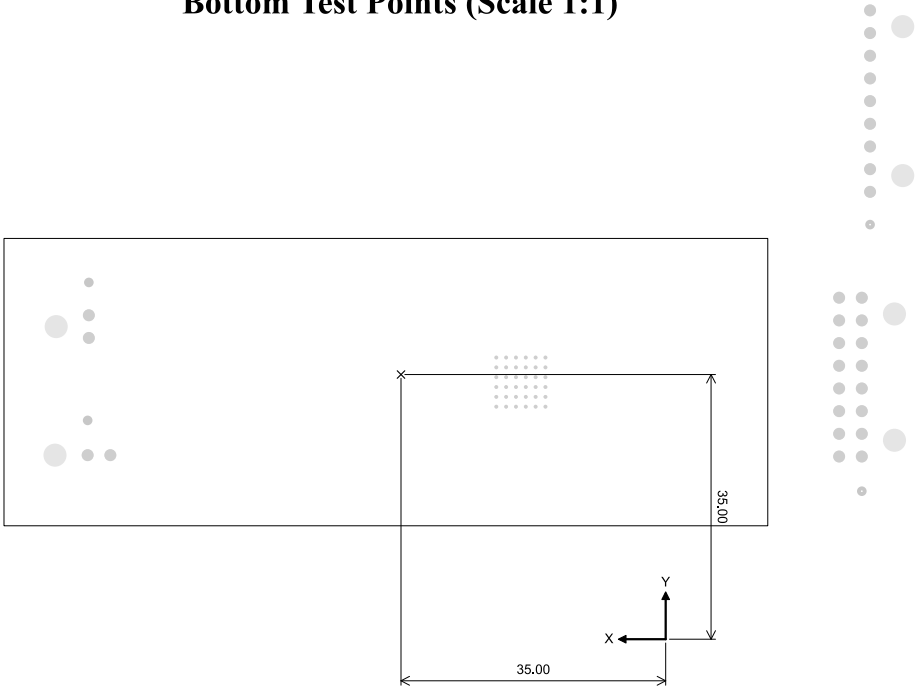
Ref.	Net	X [mm]	Y [mm]
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All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: d2b8cb1
	Sheet Title: Top Test Points (Scale 1:1)	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:	Reviewer:	Size: A4	Sheet: 5 of 10	

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Bottom Test Points (Scale 1:1)



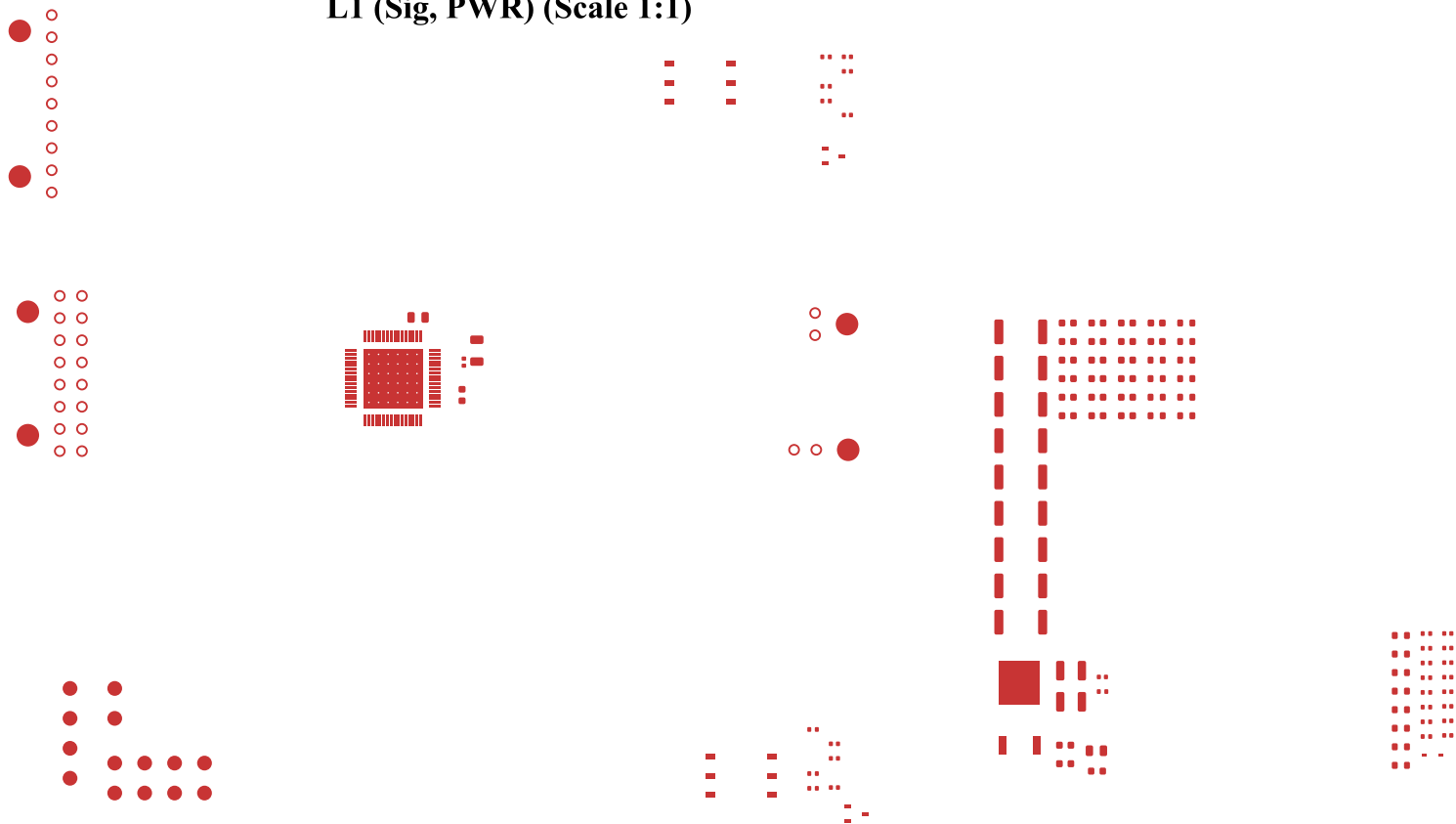
Ref.	Net	X [mm]	Y [mm]
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All dimensions are in millimeters unless otherwise specified.

	Comments:		Company:		Variant:		Git Hash:		
			Blue Devil Motor Sports		PRELIMINARY		d2b8cb1		
			Board Name:			Project Name:			
			BMS_SLAVE			BMS Slave			
Sheet Title:		File Name:		Designer:		Date:		Revision:	
Bottom Test Points (Scale 1:1)		bms_slave.kicad_pcb		Aidan Brzezinski		2024-04-13		+ (Unreleased)	
Sheet Path:				Reviewer:		Size:		Sheet:	
						A4		6 of 10	

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L1 (Sig, PWR) (Scale 1:1)



	Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: d2b8cb1
	Sheet Title: L1 (Sig, PWR) (Scale 1:1)	Board Name: BMS_SLAVE		Project Name: BMS Slave	
	Sheet Path:	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
			Reviewer:	Size: A4	Sheet: 7 of 10

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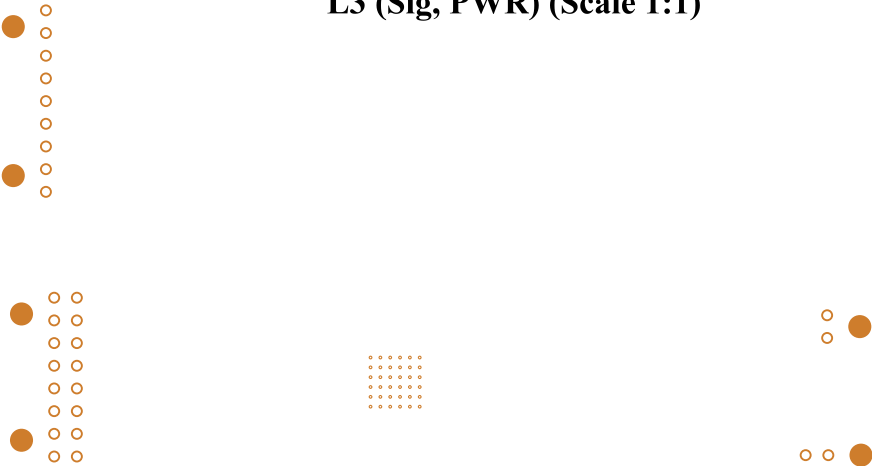
L2 (GND) (Scale 1:1)



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			Blue Devil Motor Sports		PRELIMINARY	d2b8cb1	
			Board Name:		Project Name:		
			BMS_SLAVE		BMS Slave		
Sheet Title:		File Name:		Designer:	Date:	Revision:	
L2 (GND) (Scale 1:1)		bms_slave.kicad_pcb		Aidan Brzezinski	2024-04-13	+ (Unreleased)	
Sheet Path:				Reviewer:	Size:	Sheet:	
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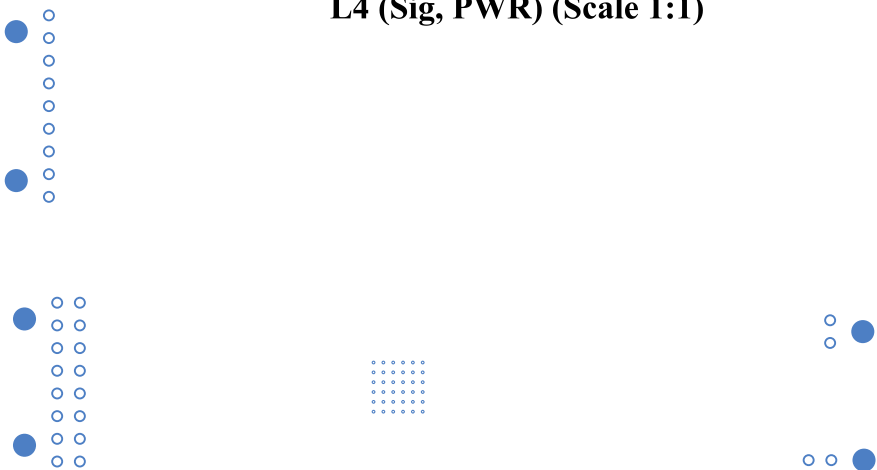
L3 (Sig, PWR) (Scale 1:1)



	Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: d2b8cb1
	Sheet Title: L3 (Sig, PWR) (Scale 1:1)	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
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L4 (Sig, PWR) (Scale 1:1)



	Comments:		Company:		Variant:		Git Hash:		
			Blue Devil Motor Sports		PRELIMINARY		d2b8cb1		
	Sheet Title:		Board Name:			Project Name:			
Sheet Path:		File Name:		Designer:		Date:		Revision:	
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Sheet Path:		Reviewer:			Size:		Sheet:		
					A4		10 of 10		